Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

R26-DS-010

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

1. Real-World Problem

Caregivers face high workload, communication barriers, delayed documentation, and patient fall risks, leading to burnout, lower care quality, and inefficient service delivery.

2. Supporting Evidence

Healthcare studies report that patient falls are a major hospital injury cause, manual documentation consumes caregiver time, and burnout negatively affects workforce performance.

3. Limitations of Existing Solutions

Current systems are mostly isolated and static, with limited integration of workforce analytics, patient monitoring, documentation automation, and communication support.

4. Proposed Solution

A multimodal computational intelligence platform integrating computer vision, speech processing, machine learning, and explainable AI to support caregivers, automate tasks, and improve decision-making.



IT22178640 | DAYANA PRIYADHARSHANI KUMAR



Component 01: AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

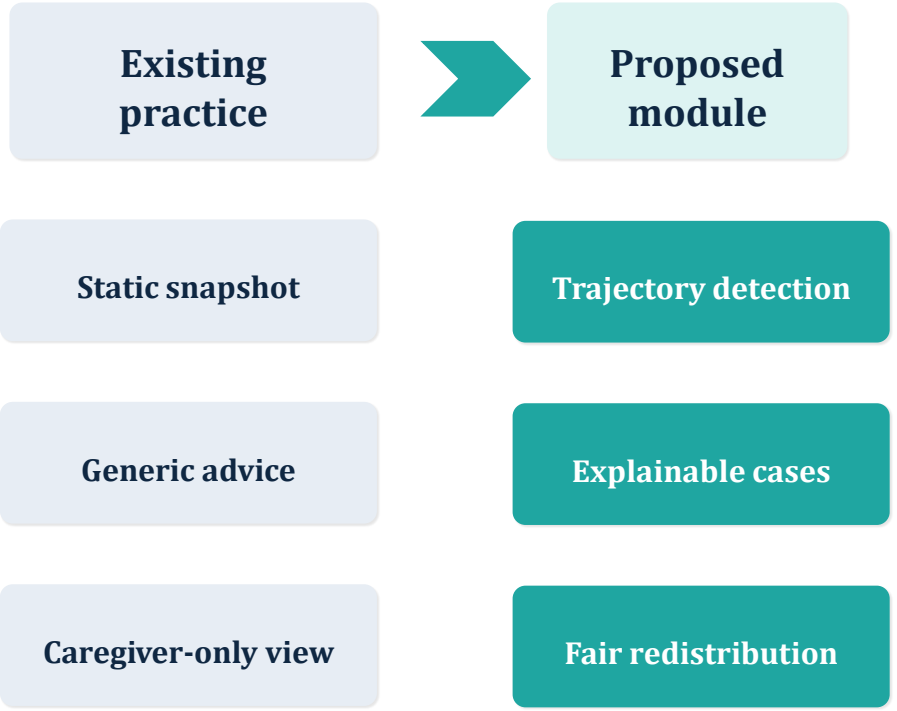
Contribution to the Overall Solution

- **Workforce layer:** Adds workforce intelligence to the overall caregiving solution.
- **Early detection:** Turns schedules, workload, events, and reflections into emerging risk signals.
- **Cross-component value:** Integrates voice-log signals and caregiver activity patterns to better understand workforce deterioration risk
- **Operational support:** Helps supervisors decide when workload should be redistributed, not only observed.
- **Explainability:** Produces case narratives that justify why a caregiver pattern is becoming unsafe.
- **System impact:** Improves caregiver performance continuity, fairness, and care-delivery stability.



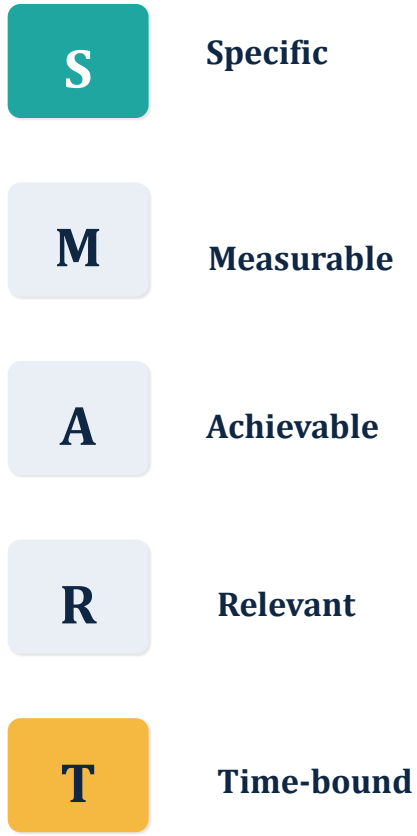
Problem Definition and Novelty

- **Core problem:** Caregiver deterioration develops gradually through unstable schedules, overload, and poor recovery.
- **Existing solutions:** Surveys, dashboards, staffing ratios, and generic wellbeing advice are usually static snapshots.
- **Main gap:** They rarely model sequence effects, team structure, or fairness across repeated assignments.
- **Novelty 1:** A dynamic personal baseline learns what is normal for each caregiver before flagging deviation.
- **Novelty 2:** Temporal risk memory distinguishes one difficult shift from persistent accumulated strain.
- **Novelty 3:** Graph-based reasoning supports explainable and fair workload redistribution instead of score-only alerts.

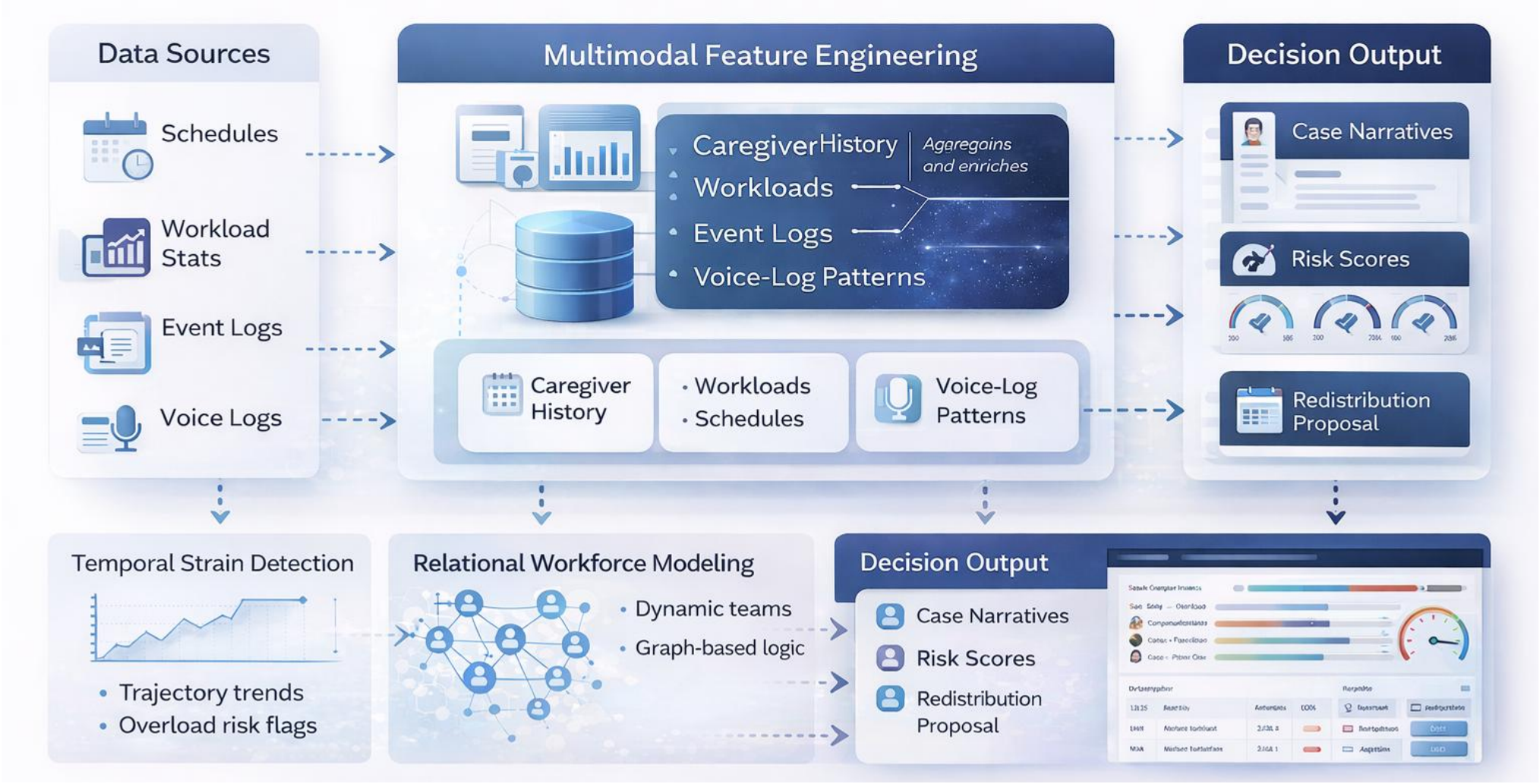


Research Objective and SMART Sub-Objectives

- **Specific:** Develop an AI-assisted module for early caregiver deterioration detection and adaptive workload redistribution.
- **Measurable:** Compute schedule stability, overload, compatibility, environmental strain, and case-priority indicators.
- **Achievable:** Prototype with operational, event, and short reflection data; validate each module on a defined workflow.
- **Relevant:** Support caregiver sustainability while helping the overall system improve care delivery quality.
- **Time-bound:** Design, implement, test, and refine the component within the project timeline using milestones.
- **Sub-objectives:** Build personal profiles, detect trajectories, discover typologies, explain cases, and suggest fair reassignments.



System Overview Diagram



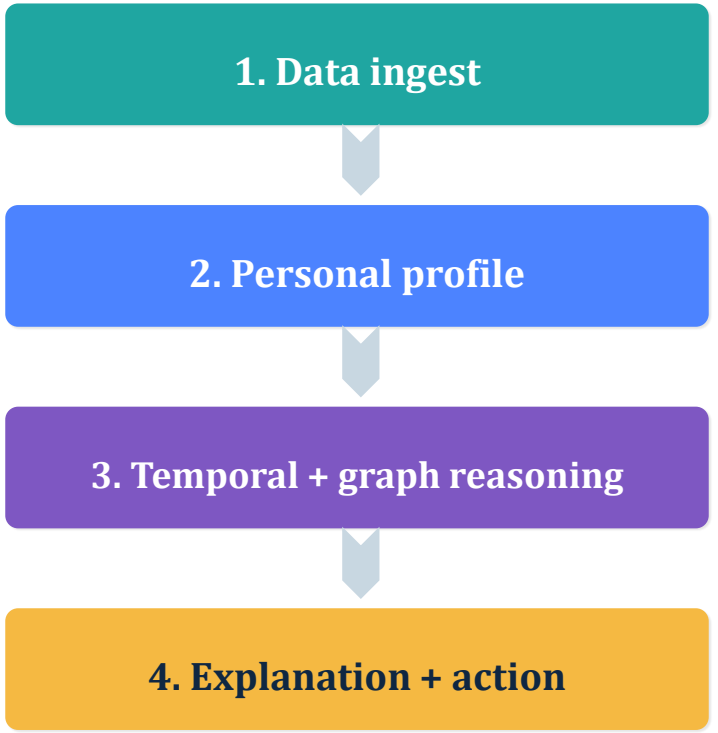
AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution



- **Domain knowledge:** Caregiver workflow, staffing fairness, patient dependency, recovery gaps, and emergency-heavy contexts.
- **Methodology base:** Multimodal feature engineering from schedules, workload, event logs, and short self-reflections.
- **Temporal layer:** Time-series modeling and accumulation/decay logic capture emerging deterioration trajectories.
- **Relational layer:** Graph modeling represents caregiver–shift–team–ward relationships for structural reasoning.
- **Latest technologies:** Graph ML, sequence learning, explainable AI, and noise-robust learning such as co-teaching.
- **Delivered output:** Soft caregiver insights plus supervisor-facing intervention cases and redistribution options.

Validation, Data Strategy, and Implementation Details

- **Validation plan:** Compare the proposed model against static scoring, simple workload rules, or non-temporal baselines.
- **Evaluation criteria:** Early-risk detection quality, explanation usefulness, fairness, and reassignment feasibility.
- **Data strategy:** Partner data is ideal; public benchmark datasets can validate sub-modules when live data is limited.
- **Security:** Use de-identification, role-based access, minimal retention, and protected storage for sensitive records.
- **Implementation flow:** Ingest data → build profile → detect risk → explain case → suggest redistribution.
- **Dependencies:** Shared authentication, common data store, and integration points with video, sign-language, and voice-logbook modules.



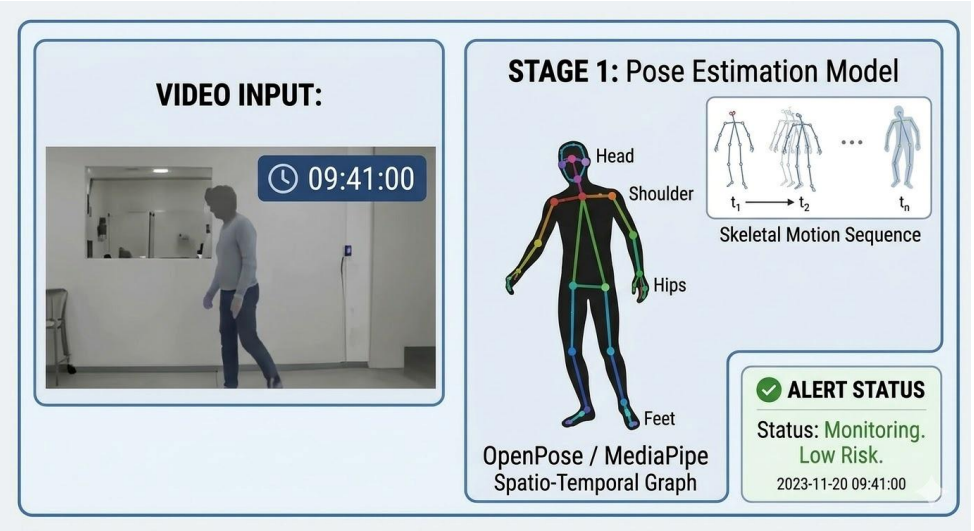
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Component 02: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction

Contribution to the Overall Solution

- **Real-time monitoring**: Continuously analyses patient movement using skeletal pose estimation to detect instability in real time.
- **Privacy-preserving analysis**: Converts video into anonymous skeletal joint data and deletes raw footage to ensure patient privacy.
- **False alarm reduction**: Differentiates normal movements from actual fall risk behaviour to minimize unnecessary alerts.
- **Immediate caregiver alerts**: Triggers multi-sensory mobile notifications (audio and vibration) when a high-risk event is predicted.
- **Explainable event visualization**: Provides a 5-second skeletal replay with an instability heatmap to help caregivers understand where the fall risk occurred.

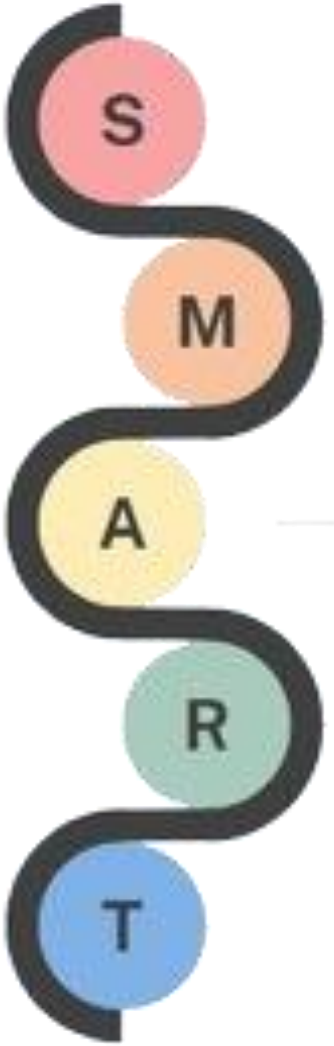


Problem Definition and Novelty

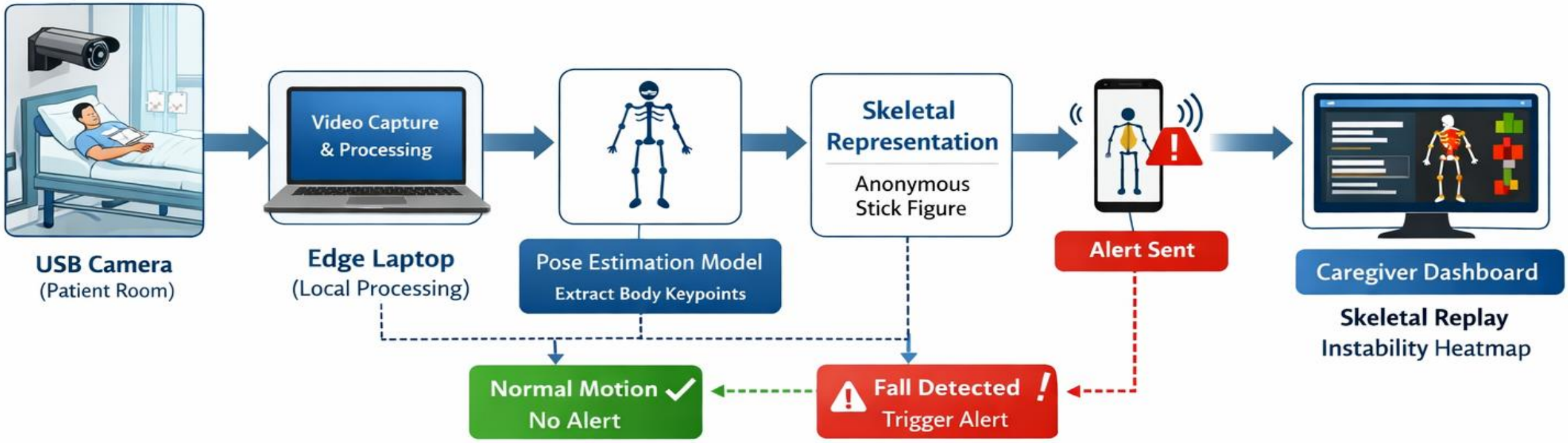
- **Core problem:** Patient falls in healthcare environments often occur suddenly and without early warning, leading to injuries and delayed caregiver response.
- **Existing solutions**: Traditional monitoring systems detect falls only after they happen or rely on video surveillance that raises privacy concerns.
- **Main gap:** Most systems cannot analyze human movement patterns over time to distinguish normal movement from real fall risk while maintaining patient privacy.
- **Novelty 1:** A privacy-preserving skeletal monitoring approach converts patient video into anonymous joint coordinates while immediately deleting raw footage.
- **Novelty 2:** Edge-based real-time analysis processes patient movement locally without cloud dependency, reducing latency and improving reliability.
- **Novelty 3:** ST-GCN-based motion prediction analyzes skeletal movement patterns to predict potential falls early and trigger caregiver alerts.

Research Objective and SMART Sub-Objectives

- **Specific**: Develop an edge-enabled skeletal movement analysis module for real-time patient fall risk prediction and caregiver alerting.
- **Measurable**: Evaluate fall prediction accuracy, false alarm rate, and response time using skeletal movement analysis.
- **Achievable**: Implement a prototype using pose estimation and ST-GCN models running on a local edge device connected to a USB camera.
- **Relevant**: Improve patient safety and caregiver response by detecting fall risks early while preserving patient privacy.
- **Time-bound**: Design, implement, test, and refine the component within the project timeline using defined development milestones.
- **Sub-objectives**: Extract skeletal pose from video, analyze motion patterns using ST-GCN, detect instability, reduce false alarms, and trigger caregiver alerts with a visual explanation.



Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction



- **USB Camera** & Edge Laptop
- **ST-GCN Model (AI)**
- Mobile Notification System
- Dashboard Interface with Heatmap

Application of Domain Knowledge and Methodology

- **Domain knowledge**: Human movement analysis, patient safety monitoring, fall risk patterns, and healthcare environment constraints.
- **Methodology base**: Skeletal pose extraction from video and feature generation from body joint coordinates.
- **Temporal layer**: Spatio-temporal motion modeling analyzes joint movement sequences to detect instability patterns.
- **Relational layer**: Skeletal graph representation models body joints and their relationships for motion understanding.
- **Latest technologies**: Pose estimation models, ST-GCN (Spatio-Temporal Graph Convolutional Networks), edge computing, and real-time computer vision.
- **Delivered output**: Real-time fall risk prediction, caregiver alerts, and privacy-safe skeletal event visualization with instability heatmap.



Validation, Data Strategy, and Implementation Details

- **Validation plan**: Evaluate the proposed model against traditional fall detection approaches and simple motion-threshold methods.
- **Evaluation criteria**: Fall prediction accuracy, false alarm rate, detection latency, and reliability of caregiver alerts.
- **Data strategy**: Public human pose datasets and simulated movement scenarios are used for model validation when real patient data is limited.
- **Security**: Ensure privacy by converting video into skeletal joint data and immediately deleting raw footage after pose extraction.
- **Implementation flow**: Capture video → extract skeletal pose → analyze motion using ST-GCN → predict fall risk → trigger caregiver alert and visualization.
- **Dependencies**: USB camera, edge laptop, pose estimation model, ST-GCN model, mobile notification service, and caregiver dashboard.

IT22294098 | VIDURSHA PRABAGARAN



Component 03: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

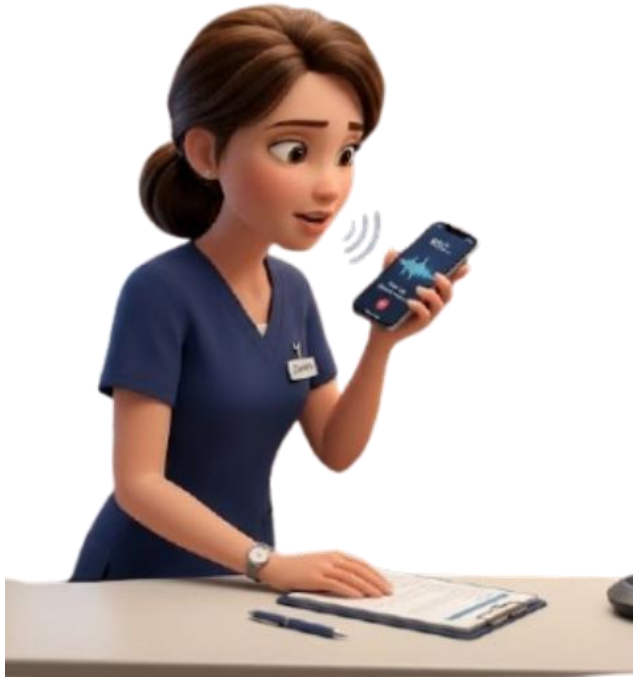
Context – Aware Intelligent System For Automated ADL Extraction And Real Time Documentation From Caregiver Voice Logs

Problem Definition:

- Manual documentation takes time and can be delayed, incomplete, or inaccurate.
- This increases caregiver workload and may reduce patient safety and care quality.
- The system converts caregiver speech into clear and structured ADL records in real time

Novelty:

- Speech recognition
- Wake-word based ambient scribing and voice shortcuts
- Auto-editing
- ADL extraction
- Intelligent reporting
- Shift handover
- 3D spoken summaries



Context – Aware Intelligent System For Automated ADL Extraction And Real Time Documentation From Caregiver Voice Logs

- **Specific:** To design and develop a context-aware intelligent system that transforms caregiver voice logs into structured ADL documentation, generates intelligent care reports, and produces automated 3D avatar-based spoken shift handover summaries.
- **Measurable:** To measure system performance using speech transcription accuracy, ADL extraction accuracy, report quality, and usability evaluation results.
- **Achievable:** To implement the system using available tools and technologies such as Whisper, spaCy, NLTK, Transformers, Porcupine and pyttsx3.
- **Relevant:** To address the real problem of time-consuming and incomplete manual documentation in caregiving environments and improve patient care quality and communication.
- **Time-bound:** To complete the design, development, implementation, and evaluation of the proposed system within the planned research period.
- **Sub-Objectives:** To analyze documentation needs, identify key ADL categories, improve caregiver speech processing, extract important care information, generate structured reports and shift handover summaries, and evaluate the effectiveness of the system.

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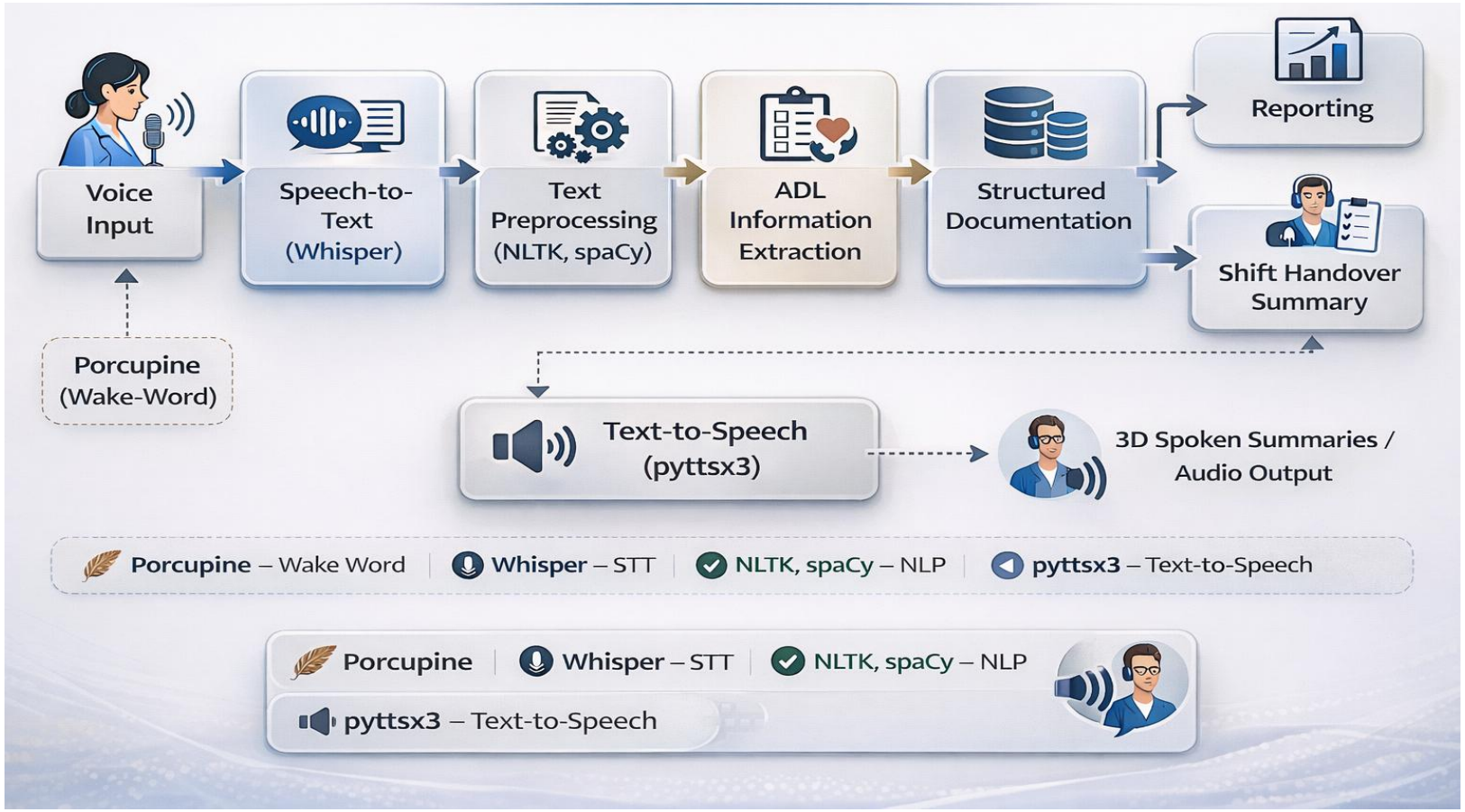
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Context – Aware Intelligent System For Automated ADL Extraction And Real Time Documentation From Caregiver Voice Logs

- **Application of Domain Knowledge:** The proposed system uses healthcare knowledge related to Activities of Daily Living such as medication, meals, mobility, hygiene, sleep, and emotional condition.
- **Research Methodology:** The study follows requirement analysis, speech processing, natural language processing, system design, implementation, and testing.
- **Knowledge Domain that Contributes to the Methodology:** The methodology is supported by healthcare knowledge, speech recognition, natural language processing, and intelligent system design.
- **Latest Technologies Used:** Whisper for speech-to-text, spaCy for NLP, NLTK for text preprocessing, Transformers from Hugging Face for context-aware summarization, Porcupine for wake-word detection, and pyttsx3 for text-to-speech.
- **Validation and Evaluation:** The system can be evaluated using transcription accuracy, ADL extraction accuracy, usability testing, and the quality of generated reports and shift handover summaries.
- **Data Availability, Security, and Ethical Clearance:** The system requires caregiver voice data and ADL-related information, while ensuring patient privacy, secure data handling, and proper ethical approval before using real healthcare data.

Context – Aware Intelligent System For Automated ADL Extraction And Real Time Documentation From Caregiver Voice Logs

System overview



Validation, Data Strategy, and Implementation Details

- **Validation:** The proposed solution will be evaluated based on speech transcription accuracy, ADL information extraction accuracy, and the quality of the generated reports and shift handover summaries.
- **Usability Testing:** The system will also be tested to measure how easy, useful, and practical it is for caregivers in real care environments.
- **Data Strategy**: The research will use caregiver voice samples and ADL-related text data for system training, testing, and performance evaluation.
- **Data Security and Ethics:** All data will be managed securely by anonymizing patient information, maintaining confidentiality, limiting access to authorized users, and following ethical guidelines during data collection and system use.



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Component 04: Personalized Sign Language Tutoring System for Caregivers

PERSONALIZED SIGN LANGUAGE TUTORING SYSTEM FOR CAREGIVERS

Problem Definition:

- Caregivers struggle to communicate effectively with deaf or hard-of-hearing care recipients during everyday care delivery.
- Current sign-learning systems are largely generic and do not adequately support caregiver-specific, guided, and adaptive learning.
- This creates a clear need for a personalized tutoring system that improves practical and confidence-building care communication.

Novelty

- Caregiver-specific sign tutoring for real care-delivery communication, instead of general sign-language learning
- Adaptive and explainable learning support through speed-tolerant evaluation and corrective feedback, not just sign recognition
- Integrated tutoring approach combining vision-based assessment, 3D avatar guidance, and personalized practice within the overall Caregiver Co-Pilot solution

PERSONALIZED SIGN LANGUAGE TUTORING SYSTEM FOR CAREGIVERS

Main Objective: To design, implement, and evaluate a personalized sign language tutoring system for caregivers that supports learning of care-related signs through speed-tolerant gesture evaluation, interpretable feedback, 3D avatar guidance, and adaptive practice.

SMART Sub-objectives

S	Define and identify care-related signs and phrases required for routine caregiving communication
M	Design and implement a computer vision-based module that evaluates caregiver-performed signs while adjusting to variation in signing speed.
A	Develop an interpretable feedback and 3D avatar guidance that supports correction of hand movement and timing errors.
R	Implement an adaptive practice mechanism that personalizes learning based on performance and repeated errors.
T	Assess the usability and effectiveness of the system through user-based evaluation and performance analysis.

Validation and evaluation plan

Validation criteria

- Technical: sign accuracy, timing tolerance, feedback consistency
- Learning: improvement before vs after tutoring
- Retention: recall after a delay / follow-up session
- Usability: clarity, confidence, usefulness, ease of learning
- Comparative baseline: static video or non-adaptive practice

Data availability

Small curated set of care-related signs; expert or reference demonstrations; learner practice recordings captured through the prototype.

Security & ethics

- informed consent for recordings
- minimal data collection and secure storage
- avoid misuse of identity-linked videos
- respect Deaf culture and non-replacement of interpreters
- institutional ethical clearance before user studies

Knowledge domains contributing to the methodology

Caregiving & health communication
selects care-relevant signs/phrases and realistic communication scenarios

Computer vision & HCI
captures hand/body motion and supports real-time learner interaction

Spatiotemporal gesture analysis
compares signs fairly despite speed variation

Explainable AI & learning science
turns errors into simple, actionable teaching feedback

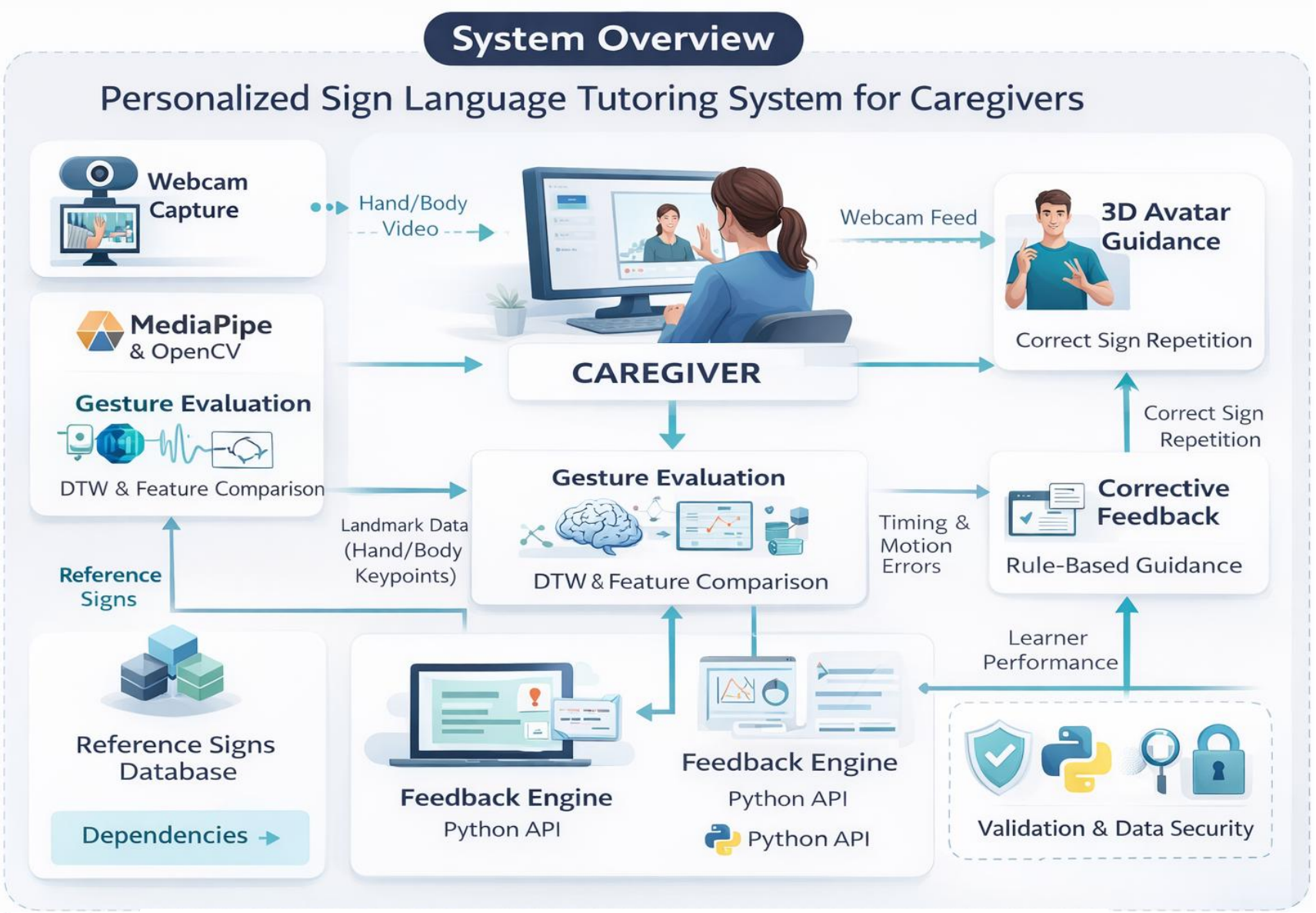
Adaptive tutoring
repeats and reinforces signs based on learner weaknesses

Latest technologies used

- Webcam-based hand/body landmark extraction
- Skeleton and spatiotemporal gesture modeling
- Pose-aware feedback generation
- 3D avatar visual guidance
- Adaptive learner progress tracking
- Web-based deployment for accessibility

Methodology = domain knowledge + AI tutoring logic

PERSONALIZED SIGN LANGUAGE TUTORING SYSTEM FOR CAREGIVERS

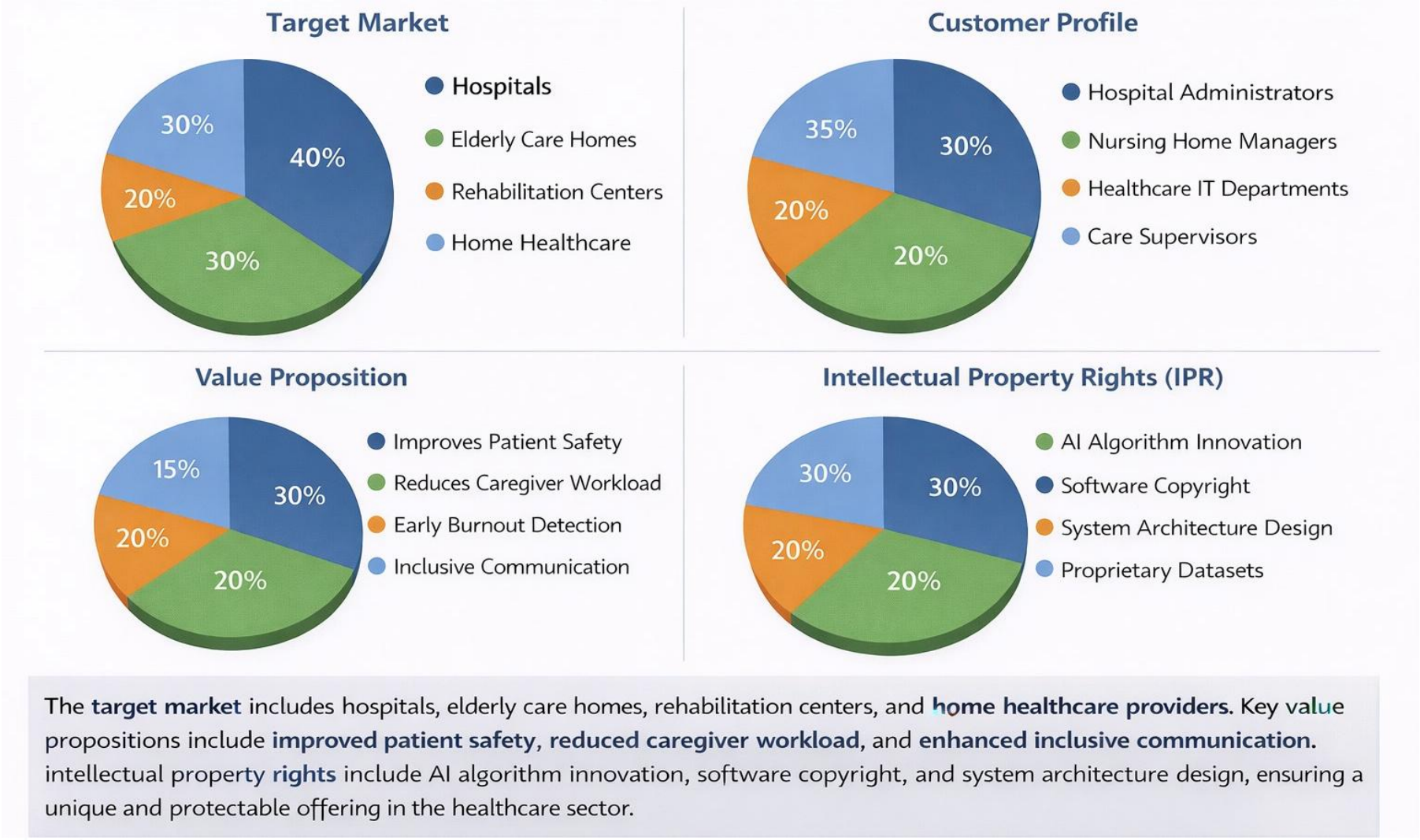


Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

Ability of Commercialization – Key Points

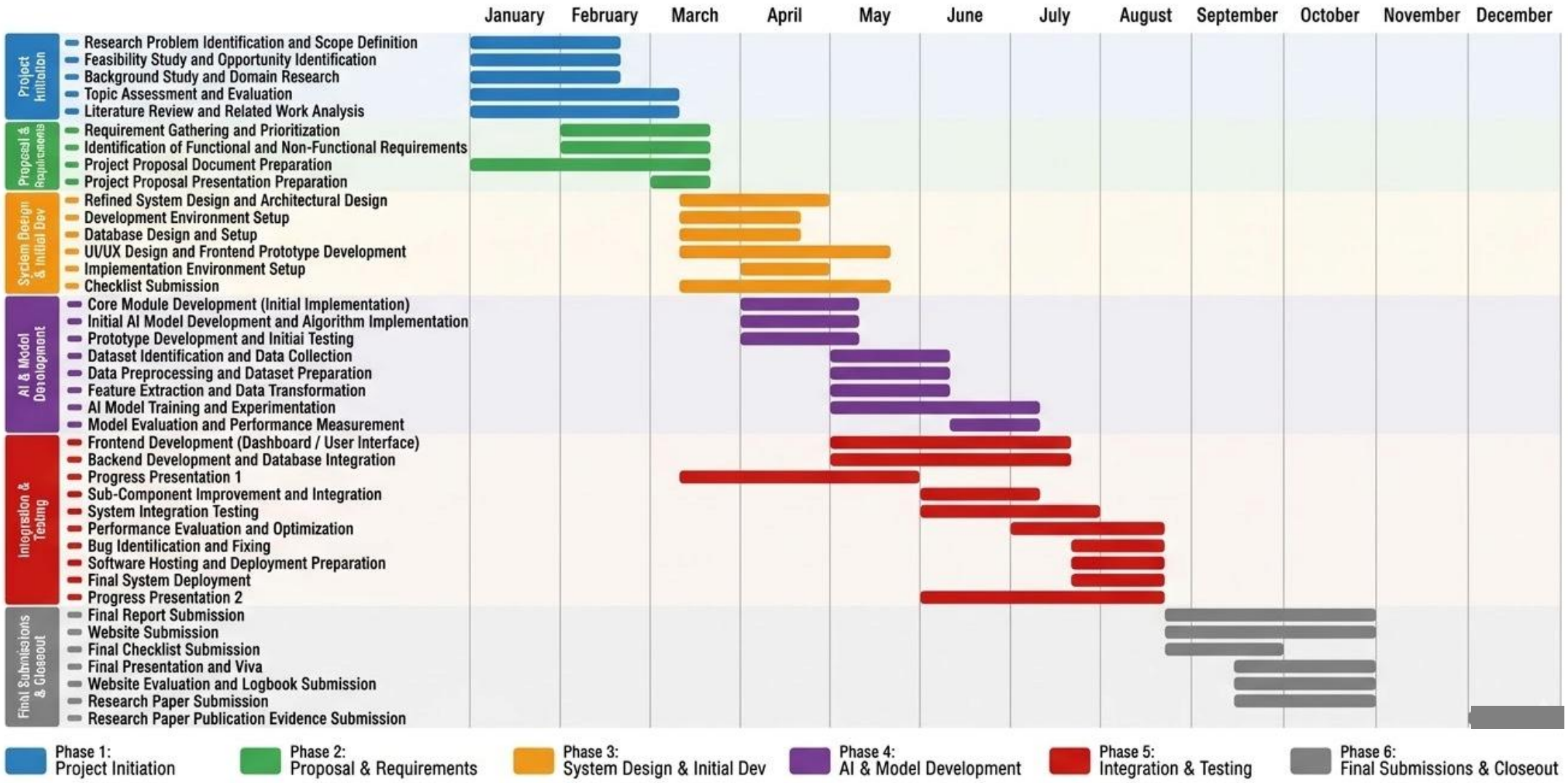
Key Point	Description	
Integrated AI Healthcare Platform	The project can be commercialized as a unified AI-powered caregiver support platform combining monitoring, documentation, workforce analytics, and communication tools.	
High Demand in Healthcare Sector	Hospitals, elderly care homes, and rehabilitation centers face increasing demand for digital solutions to improve patient safety and caregiver efficiency.	
Improves Patient Safety	The real-time fall prediction system can significantly reduce patient injuries and liability risks, making it valuable for healthcare institutions.	
Reduces Caregiver Workload	Automated voice-based documentation and workload monitoring reduce administrative burden and improve caregiver productivity.	
Workforce Management Optimization	The caregiver deterioration detection module helps healthcare organizations optimize staff allocation and prevent burnout.	
Supports Incllisive Healthcare Communication	The sign language tutoring system enables caregivers to communicate effectively with deaf or hard-of-hearing patients.	**Privacy-Preserving Monitoring Technology** The skeletal pose-based fall detection ensures patient privac, increasing acceptance in healthcare environments with strict data protection requirements.
Scalable Deployment Model	The platform can be deployed across multiple healthcare facilities through cloud or edge-based infrastructure.	**Multiple Revenue Opportunities** The solution can be commercialized through SaaS subscriptions, enterprise licensing, or healthcare technology partnerships.

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery



The **target market** includes hospitals, elderly care homes, rehabilitation centers, and **home healthcare providers.** Key value propositions include **improved patient safety, reduced caregiver workload,** and **enhanced inclusive communication.** intellectual property **rights** include AI algorithm innovation, software copyright, and system architecture design, ensuring a unique and protectable offering in the healthcare sector.

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery – Gantt Chart



Work Breakdown Structure

Multimodal Computational Intelligence for Enhancing ***Caregiver Performance*** *and* ***Care Delivery***

